COURSE CODE/TITLE: CPE 031 - Visualizations and Data Analysis	**SECTION: CPE21S1**
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ASSIGNMENT NO. 10.1 The Power in Effective Data Storytelling

1. PROCEDURE OUTPUT

What procedures did you perform?

Guide Questions:

1. What is the "power" in data storytelling that the speaker emphasizes, and why does she believe it matters in influencing decisions or actions?
2. How does she define or illustrate the role of audience awareness in crafting a data story?
3. What example(s) does she give of a visualization or story gone wrong and what are the root causes of the failure according to her?
4. Which "preattentive" or design principles does she highlight as important for making charts communicate more clearly (for example: color, size, position, emphasis)?
5. Based on her advice, how would you redesign one of your own visualizations (or a chart you've seen) to better tell a story? What would you change and why?

1.) The **POWER** that the speaker is emphasizing is the power or how the power of data storytelling can show or uncover the kind of a "Plot Twist" which basically changes the narrative that the reader might initially have at first. This matters because we as a society, or all the current human data production as of today produces large and large amounts of data, but allegedly, only 0.5% of all the data was used. And due to this extra large amount of data, a user or a data analyst or like that will use this data without a goal would be swamped and will take a long long time to make a decision or take action to do something.
2.) The speaker defined the audience awareness as the important practice of knowing precisely the problem a decision maker needs to resolve before opening or finding a dataset and creating graphs. A skilled data storyteller makes the narrative directly told to the audience's objectives, wants, or requirements. She illustrates this principle through the commercial failure of the 1992 IBM Simon, which attempted to combine multiple features rather than focus on what the market wants, resulting in an expensive device that lacks the ability to be used like a modern mobile phone.
3.) To illustrate visualization and the failures, the speaker emphasized the commercial collapse of the IBM Simon, and the aimless analytical routine where the analysts spend late night staring helplessly at spreadsheets while drinking coffee. According to her, the primary root cause of all these failures is the "jumping into data analyst without defining a core problem" . A second cause is the absence of a clear game plan, which leaves the analysts without a structured set of guiding questions to filter their findings toward a meaningful resolution. Lastly, is the uncritical acceptance of flawed raw data, because it produces misleading visual or invalid conclusions regardless of how grand the chart may appear.
4.) She emphasizes that the preattentive design principles can help reduce cognitive load and enhance clarity. The speaker advocates maintaining a singular narrative focus within a chart to prevent cognitive fatigue caused by competing trends. Lastly, she emphasizes that using visual contrast such as

selective color and callouts to direct the audience's eye immediately to the plot twist, while emphasizing that clear visual design relies fundamentally on high quality but verified and accurate data.

5.) Well, I can apply Sridhar's advice by redesigning a cluttered multi category chart with four lines or columns, to a single one by filtering the others that are not the focus into a single one. I can frame the next agenda where I can sort out the unnecessary parts to make the data at the chart be more focused on a certain question. Then from the original generic title, I could change the title to a focus to the subject one, and adding the information that I have gathered, to add a narrative and preattentive attribute to the storytelling that I have.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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